

FREDERICTON CDA, NB, Canada

WMO: 716680

Lat: 45.9203N

Lon: 66.6089W

Elev: 115

StdP: 14.63

Time Zone: -4.00 (NAA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-3.2	-16.3	2.3	-6.1	-11.9	2.9	-1.6	24.8	23.2	22.5	22.5	5.4	290	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.7	69.3	81.8	68.0	79.1	66.2	72.0	80.7	70.3	78.4	68.7	75.9	9.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	107.7	76.3	67.5	101.5	74.2	66.0	96.3	72.4	35.8	80.9	34.3	78.8	33.0	75.9	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.6	16.4	-15.6	90.1	6.0	2.3	-19.9	91.8	-23.4	93.1	-26.7	94.4	-31.1	96.1		
			-16.0	74.8	5.8	1.5	-20.2	75.9	-23.6	76.8	-26.8	77.7	-31.0	78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	18.0	19.0	28.2	41.2	52.9	61.2	68.5	66.5	59.3	47.7	36.2	24.2
	DBStd	19.39	11.75	10.62	9.61	7.04	6.43	5.41	4.65	4.60	6.88	7.27	8.64	10.12
	HDD50	4208	993	867	677	276	39	1	0	0	6	129	421	799
	HDD65	8033	1458	1287	1140	715	380	137	20	35	197	535	865	1264
	CDD50	1913	0	0	2	11	128	337	573	513	284	59	6	0
	CDD65	264	0	0	0	0	4	23	128	83	25	1	0	0
	CDH74	2116	0	0	3	6	101	230	969	605	195	7	0	0
	CDH80	447	0	0	0	1	22	42	222	113	47	0	0	0
	WSAvg	7.2	8.0	8.3	8.7	8.1	7.3	6.6	6.1	5.7	6.1	7.0	7.5	7.6
	PrecAvg	45.3	4.1	3.4	3.8	3.4	3.8	3.3	3.5	3.6	3.7	3.8	4.4	4.6
Precipitation	PrecMax	59.8	8.8	6.5	7.6	6.1	8.3	7.8	6.9	9.4	10.7	7.2	8.1	8.1
	PrecMin	33.3	0.8	1.1	1.1	0.8	0.8	1.0	1.1	0.7	0.8	0.8	1.7	1.9
	PrecStd	6.4	1.7	1.4	1.4	1.3	1.9	1.6	1.4	2.0	1.8	1.7	1.5	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	48.4	61.4	71.1	82.6	84.9	89.0	86.9	86.1	73.8	62.9	53.0
		MCWB	49.9	46.4	50.6	53.2	65.5	66.3	71.2	71.6	70.5	61.1	60.1	51.3
	2%	DB	44.0	41.3	49.3	63.6	76.1	79.9	85.1	83.1	79.3	67.2	57.9	46.7
		MCWB	42.4	36.9	40.7	50.9	61.1	64.8	69.7	69.3	67.4	59.8	54.0	45.1
	5%	DB	38.7	37.6	44.7	58.4	70.7	76.1	82.3	80.1	74.8	63.3	54.0	41.8
		MCWB	36.5	34.6	39.2	48.5	57.1	62.9	68.6	67.4	65.1	58.3	50.4	39.7
	10%	DB	34.4	34.3	40.9	53.8	66.2	72.9	79.6	77.3	71.0	60.0	49.6	37.9
		MCWB	32.5	32.1	35.6	46.0	54.7	61.3	67.3	66.1	63.3	54.9	46.5	35.9
	0.4%	WB	50.4	46.2	52.1	58.6	67.4	70.6	74.3	74.2	72.0	66.1	60.8	51.4
		MCDB	51.4	48.6	58.6	65.7	79.6	79.1	84.1	83.0	82.6	68.6	62.2	52.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.6	38.4	43.3	54.2	62.9	68.2	72.4	71.8	69.4	62.1	55.5	45.3
		MCDB	43.7	40.2	46.2	59.1	72.2	75.6	81.2	79.7	76.4	65.2	57.3	46.6
	5%	WB	36.6	34.7	39.6	50.3	59.6	65.7	70.7	69.8	67.1	59.2	51.2	40.2
		MCDB	38.3	36.9	43.8	56.7	67.1	72.3	79.0	77.3	72.6	63.3	53.4	41.5
	10%	WB	32.7	32.3	36.4	46.5	56.6	63.4	69.0	68.0	64.8	55.4	46.3	36.0
		MCDB	34.4	34.2	40.5	52.7	63.6	69.9	76.8	74.2	69.4	58.8	49.3	37.5
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.5	16.9	18.0	20.1	18.6	19.0	18.9	19.2	16.0	13.7	13.4
		MCDBR	20.9	18.1	20.7	27.8	29.2	25.3	23.4	23.5	22.9	21.2	19.9	17.0
		MCWBR	20.3	15.5	15.1	16.9	15.6	13.1	11.0	11.7	13.0	14.8	16.7	16.8
	5% WB	MCDBR	20.9	17.2	19.3	23.0	24.4	20.9	20.0	19.9	19.9	18.3	17.4	17.1
		MCWBR	20.6	15.6	15.8	17.0	15.2	12.9	10.6	11.4	12.9	15.3	16.8	17.3
Clear-Sky Solar Irradiance	taub		0.274	0.285	0.316	0.346	0.373	0.405	0.424	0.398	0.369	0.327	0.305	0.286
	taud		2.456	2.407	2.432	2.425	2.392	2.306	2.274	2.347	2.405	2.527	2.540	2.477
	Ebn at Noon		268	286	291	290	284	274	266	269	268	267	254	250
	Edh at Noon		21	27	31	34	36	40	41	37	32	24	20	19
All-Sky Solar Radiation	RadAvg		474	750	1114	1381	1583	1712	1740	1540	1244	768	450	351
	RadStd		50	68	73	129	181	130	138	98	91	73	52	33

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	N/A

Nomenclature: See separate page